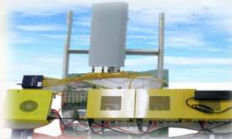




GENI in the Classroom



GENI and NDN

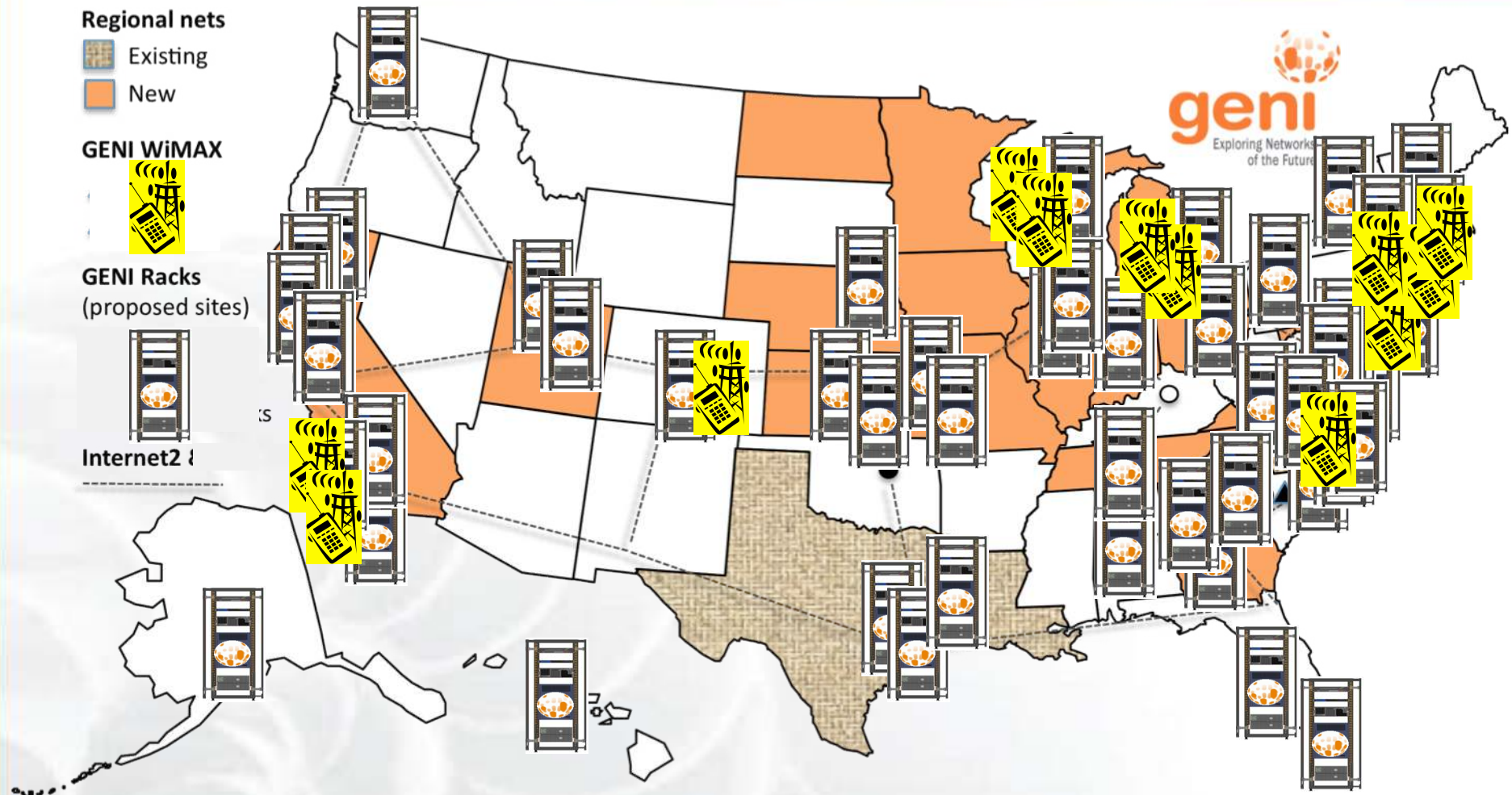
... or why should I use GENI?

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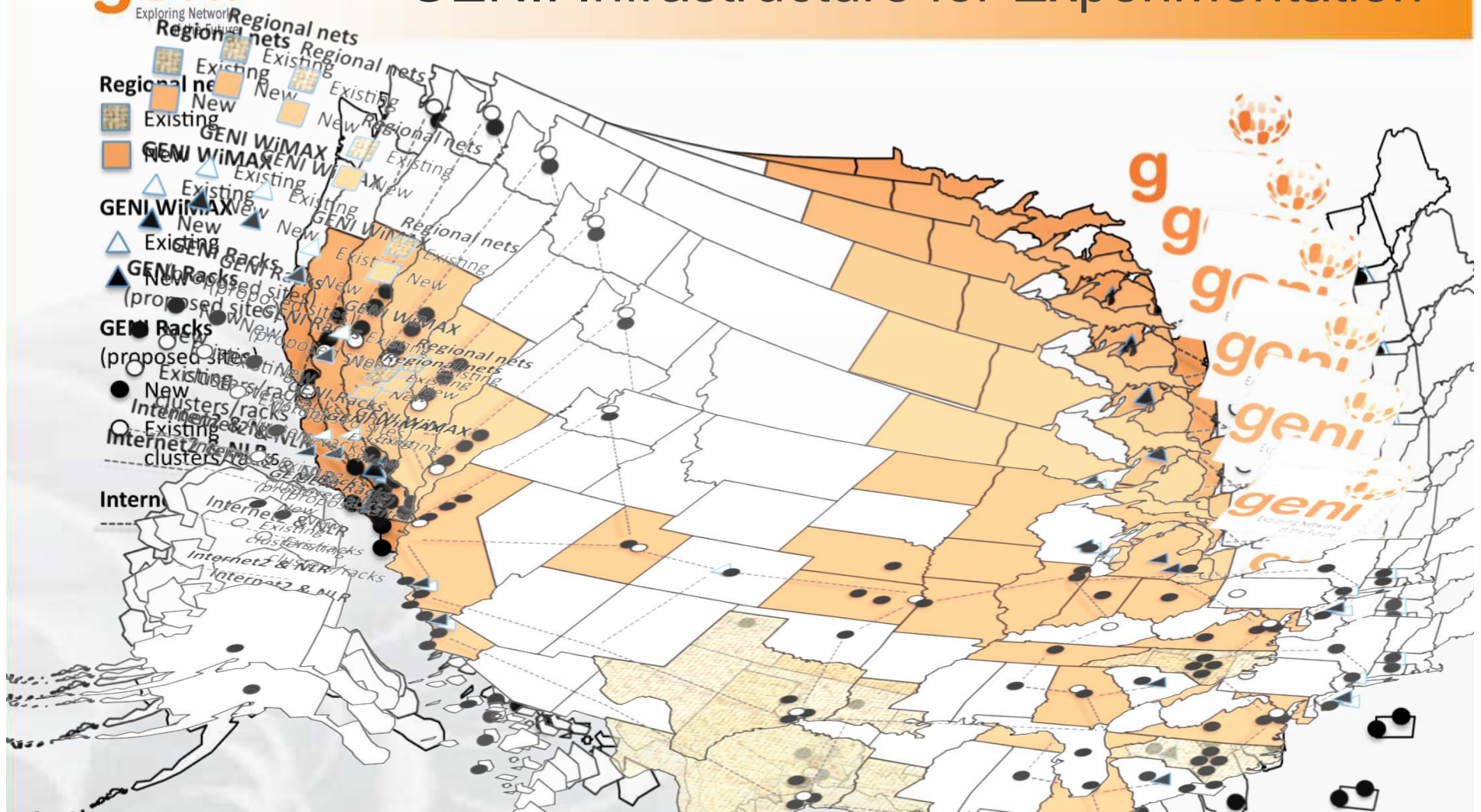
GENI: Infrastructure for Experimentation



GENI provides compute resources that can be connected in experimenter specified Layer 2 topologies.



GENI: Infrastructure for Experimentation

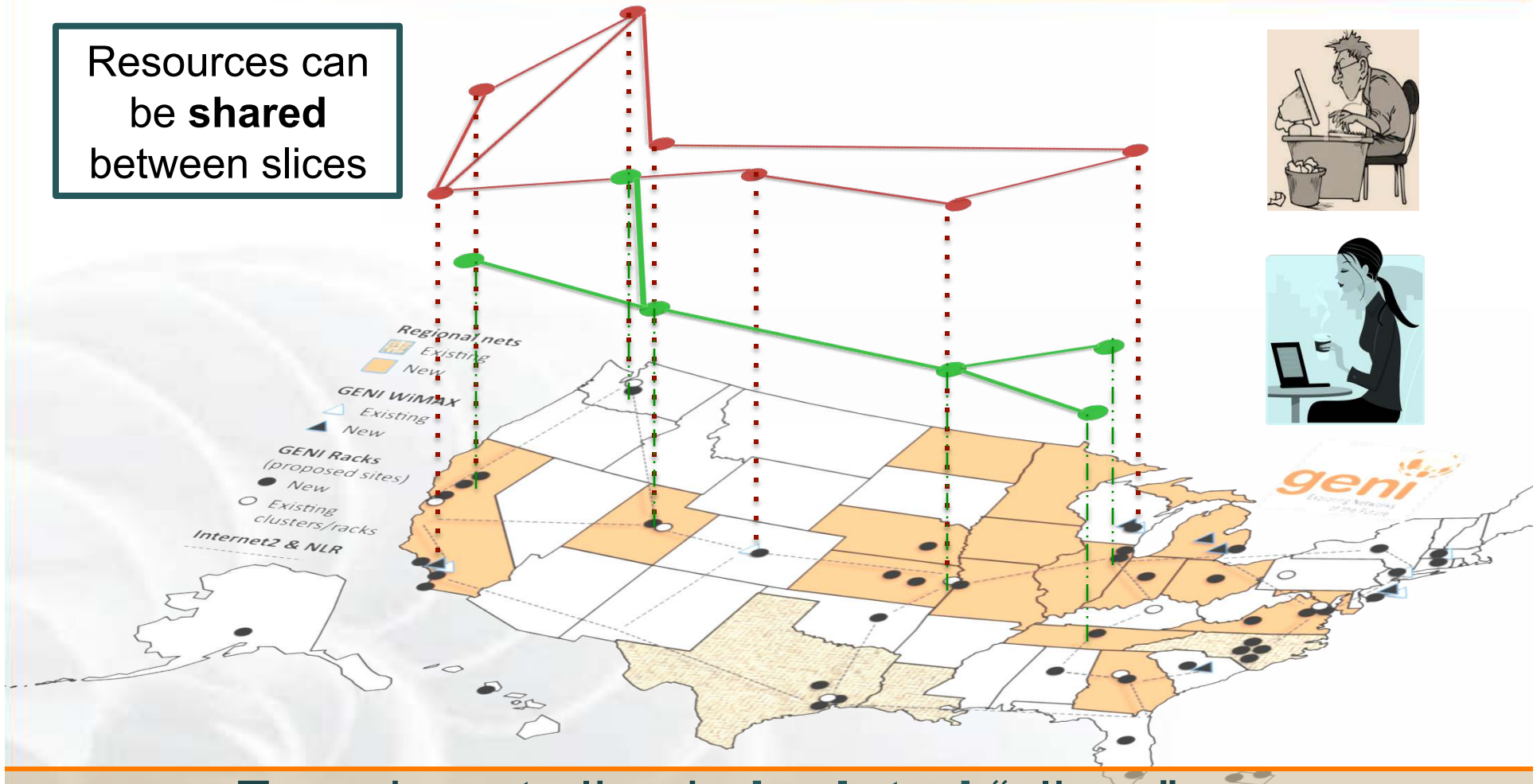


GENI provides compute resources that can be connected in experimenter specified Layer 2 topologies.



Multiple GENI Experiments run Concurrently

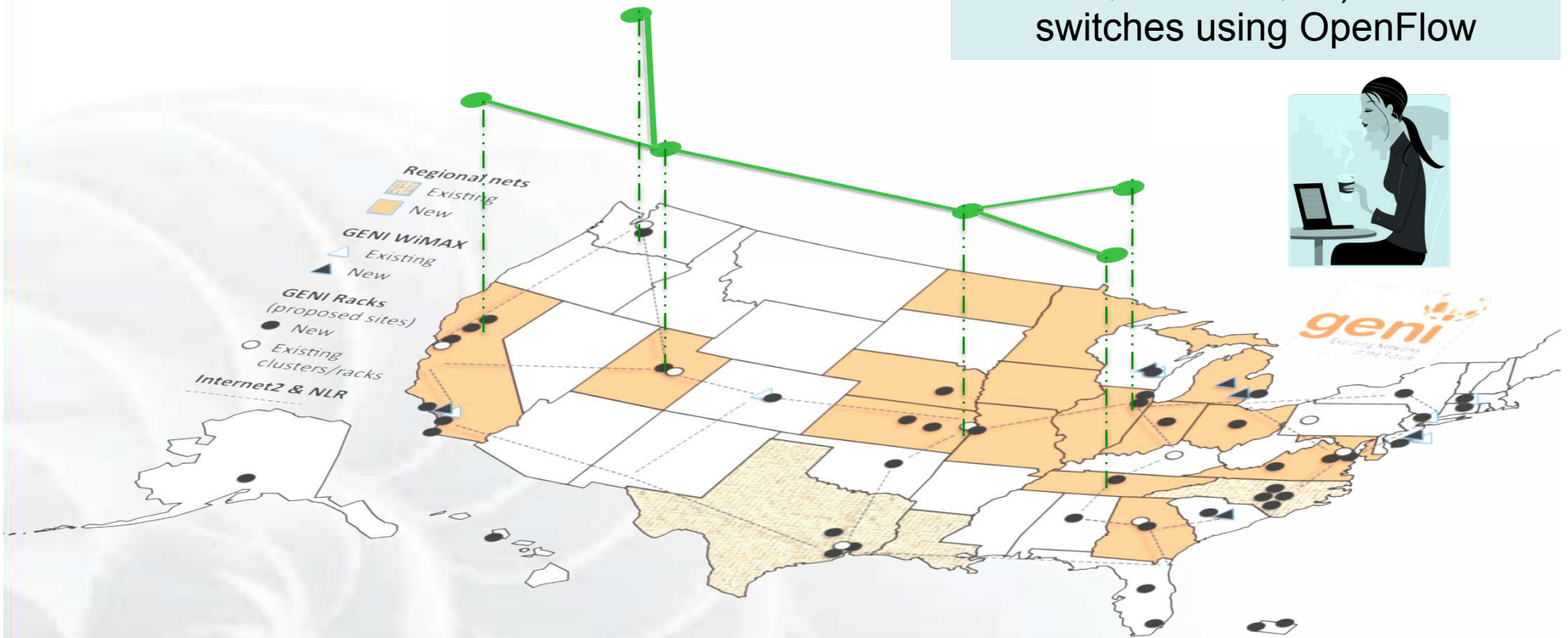
Resources can
be **shared**
between slices



Experiments live in **isolated “slices”**
over 2300 users, dozen classes per semester

GENI is “Deeply Programmable”

I install software I want
throughout my network slice (into
routers, switches, ...) or control
switches using OpenFlow

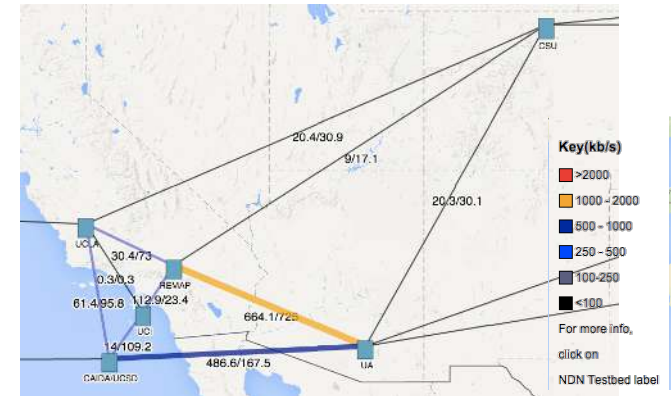


Experimenters can set up custom topologies, protocols and
switching of flows

- Main design principles
 1. **Sliceable**: provide isolated sandboxes
 2. **Deeply programmable**: compute and storage in the network
- Wide Area Layer 2 networks
- Fine grained control over topology design
 - Geographical locations, size and type of topologies
- Tools for experimentation
 - Orchestrate large deployments, monitor, archive, automate
- Enables collaboration
 - Virtual lab
 - Easy to share experiment configurations

NDN-specific testbeds

- Production prototype NDN network, centrally managed
- ONL: single site programmable testbed



How can GENI help expand your testbed?

1. Expand the core using nodes in GENI

- Richer topology
- Easy to bring up edge nodes
- Layer 2 connectivity (multipoint AL2S VLAN)

2. Use as a sandbox

- Easy to bring up multi-site private NDN networks
- Experiment with wireless, SDN
- Experiment with L2

3. Training

- Tutorials, online available to all GENI Users
- Classes

Outsource authentication, easier for new researchers to get up and running

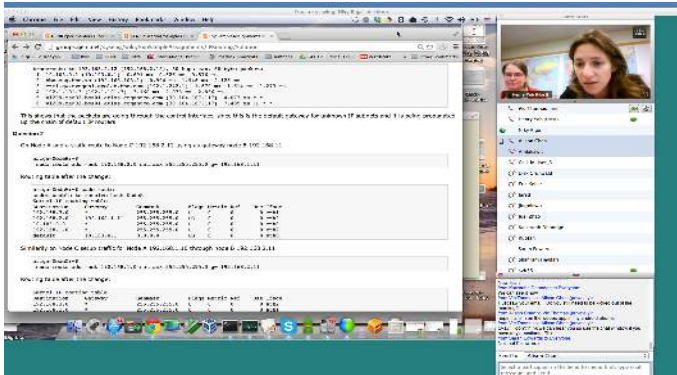
- Wide Area Layer 2 network
 - Extensive control over the network, geographic locations, isolated VLANs
- GENI provides diverse resources
 - Compute, storage, raw pcs, vm servers
 - Wireless infrastructure
 - Programmable switches
- Free for research and education

GENI's International Collaborations



GENI is working actively with peer efforts on five continents to define and adopt common concepts and APIs.

Upcoming GENI Events



Train-the-TA (Sep 11th – 18th)
Offered online at the start of
each semester

GEC21



Bloomington, Indiana

October 20-23, 2014

GENI Engineering Conferences, held three times a year

Planning & discussion for experimenters, software, infrastructure

Tutorials and workshops

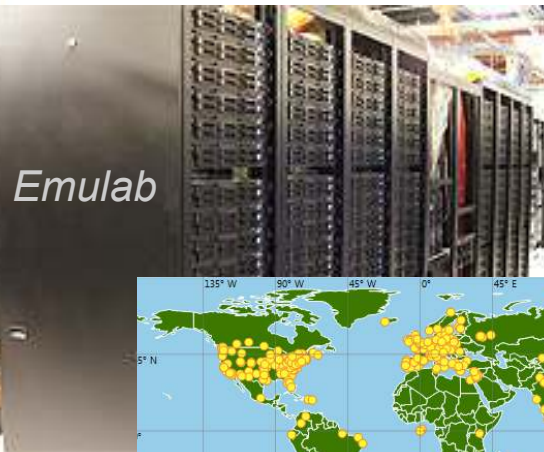
Travel grants to US academics for participant diversity

GEC22: Special event at Washington, March 2015

- Invitees to include **leaders from Government agencies, companies** and your peers from academia and industry
- Plenary demos to be one of the best show and tell performances
- A great **opportunity** to communicate your message and have maximum impact

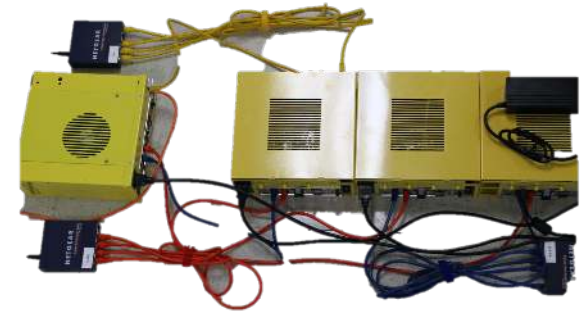
THANK YOU!

GENI Compute Resources



Existing Testbeds

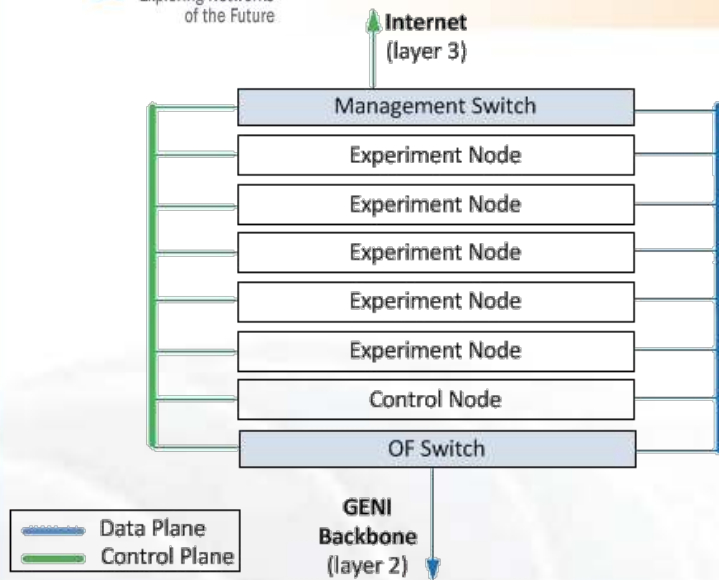
GENI Racks



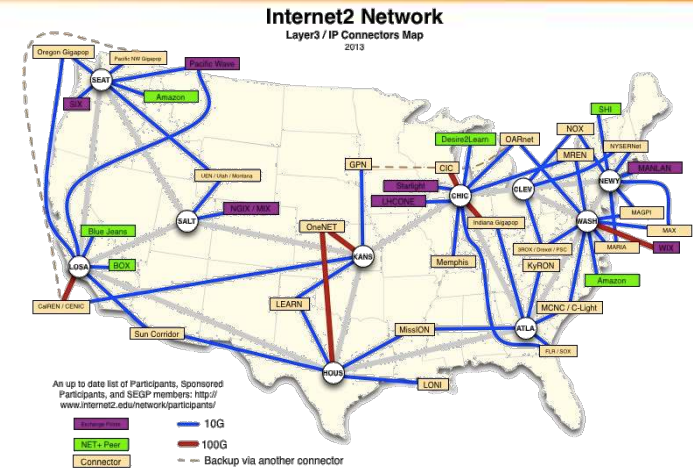
GENI Wireless
compute nodes



GENI Networking Resources

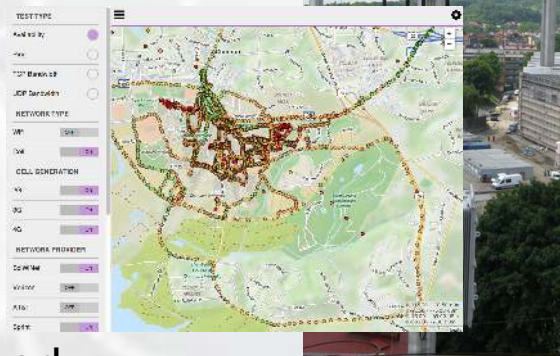


Networking within a Rack



National Research Backbones
(e.g. Internet2)

4G/3G
GENI network



WiMAX Base Stations



Regional Networks
(e.g. CENIC)